INT306:DATABASE MANAGEMENT SYSTEMS

L:3 T:0 P:2 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: explain the purpose, components, and architecture of database systems, and describe data modeling concepts including ER models and relational models.

CO2 :: demonstrate SQL data definition, manipulation, control, and transaction commands along with database keys and integrity constraints.

CO3 :: apply relational operations including joins, aggregate functions, subqueries, views, window functions, and indexing concepts to retrieve and organize data efficiently.

CO4 :: analyze relational database schemas using functional dependencies and normalization techniques to identify and reduce data redundancy and anomalies.

CO5 :: determining database programming constructs including flow control statements, functions, stored procedures, triggers, cursors, and transaction processing concepts.

CO6 :: outline the concepts and characteristics of NoSQL databases including MongoDB, JSON databases, serverless databases, and vector databases.

Unit I

Introduction to database : purpose of database systems, components of dbms, applications of dbms, three tier dbms architecture, data independence, database schema, instance, data modeling, entity relationship model, relational model, Comparison of relational and non-relational databases

Unit II

Relational query language : introduction to data definition language, data manipulation, data control and transaction control language, integrity constraints, database keys, SQL basic operations

Unit III

Relational Operations : Aggregate functions, Sql joins, set operators, views, subqueries, relational algebra, Window functions, Hashing, Indexing and Query Optimization.

Unit IV

Relational Database Design : data integrity rules, functional dependency, need of normalization, first normal form, second normal form, third normal form, boyce codd normal form, multivalued dependencies, fourth normal form

Unit V

Programming Constructs in Databases : flow control statements, functions, stored procedures, cursors, triggers, exception handling

Database Transaction Processing : transaction system concepts, desirable properties of transactions, schedules, serializability of schedules, concurrency control, recoverability

Unit VI

NoSQLDatabases : Introduction of MongoDB, DynamoDB, Serverless cloud database, Structure of MongoDB, SQL vs NoSql, Working with MongoDB, JSON databases, JSON representation of part of the dataset, Index creation & performance comparison using EXPLAIN, Vector Databases

List of Practicals / Experiments:

Introduction to Data Definition Language and Managing Schema Objects

- Data Definition Language
- Categorize Database Objects
- Create Tables
- Describe the data types
- Understand and Manage Constraints
- Create a table using a subquery
- How to alter a table?
- How to drop a table?

- Creating and using temporary tables
- Creating and using external tables

Data Manipulation

- Add New Rows to a Table
- Change the Data in a Table
- Use the DELETE and TRUNCATE Statements
- How to save and discard changes with the COMMIT and ROLLBACK statements
- Implement Read Consistency
- Describe the FOR UPDATE Clause

Aggregated Data Using the Group Functions

- Usage of the aggregation functions in SELECT statements to produce meaningful reports
- Describe the AVG, SUM, MIN, and MAX function
- How to handle Null Values in a group function?
- Exclude groups of date by using the HAVING clause

Display Data from Multiple Tables

- Write SELECT statements to access data from more than one table
- Natural Join
- Non equijoins
- View data that does not meet a join condition by using outer joins
- Join a table to itself by using a self join
- Create Cross Joins

SET Operators

- Describe the SET operators
- Use a SET operator to combine multiple queries into a single query
- Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators
- Matching the SELECT statements
- Use the ORDER BY Clause in Set Operations

Creating Views

- Create, modify, and retrieve data from a view
- Perform Data manipulation language (DML) operations on a view
- How to drop a view?

Introduction to PL/SQL

- PL/SQL Overview
- List the benefits of PL/SQL Subprograms
- Overview of the Types of PL/SQL blocks
- Create a Simple Anonymous Block
- Generate the Output from a PL/SQL Block

PL/SQL Identifiers

- List the different Types of Identifiers in a PL/SQL subprogram
- Use of variables to store data
- %TYPE Attribute

Explicit Cursors

- Understand Explicit Cursors
- Declare the Cursor
- How to open the Cursor?
- Fetching data from the Cursor
- How to close the Cursor?
- Explicit Cursor Attributes

Exception Handling

- What are Exceptions?
- Handle Exceptions with PL/SQL
- Trap User-Defined Exceptions
- RAISE_APPLICATION_ERROR Procedure

Stored Procedures and Functions

- What are Stored Procedures and Functions?
- Differentiate between anonymous blocks and subprograms
- Create a Simple Procedure
- Create a Simple Procedure with IN parameter
- Create a Simple Function
- Execute a Simple Procedure
- Execute a Simple Function

Relational Query Languages

- relational algebra
- introduction to data definition language, data manipulation
- data control and transaction control language
- integrity constraints
- database keys
- sql basic operations
- aggregate functions
- sql joins
- set operators, views
- subqueries

Programming constructs in Database

- flow control statements
- functions
- stored procedures
- cursors
- triggers
- exception handling

Text Books:

1. POSTGRESQL: UP AND RUNNING - A PRACTICAL GUIDE TO THE ADVANCED OPEN SOURCE DATABASE, THIRD EDITION by REGINA OBE, LEO HSU, SHROFF/O'REILLY

References:

1. DATABASE SYSTEM CONCEPTS by HENRY F. KORTH, ABRAHAM SILBERSCHATZ, S. SUDARSHAN, MCGRAW HILL EDUCATION

References: 2. SIMPLIFIED APPROACH TO DBMS by PRATEEK BHATIA AND GURVINDER SINGH, KALYANI PUBLISHERS